

Node + Express → GitHub → Render Deployment Complete Setup From Scratch

This guide explains the full workflow: creating a backend locally, using environment variables, pushing to GitHub, and deploying to Render. Every step includes commands, purpose, and sequence.

PART 1 — Create Backend Locally

Step 1: Create project folder

```
mkdir back_ptc_1  
cd back_ptc_1
```

Step 2: Initialize Node project

```
npm init -y
```

Step 3: Install Express

```
npm install express
```

Step 4: Install nodemon (optional)

```
npm install nodemon --save-dev
```

Step 5: Install dotenv

```
npm install dotenv
```

PART 2 — Create Files

index.js

```
require('dotenv').config()
const express = require('express')
const app = express()

const port = process.env.PORT || 3000

app.get('/', (req, res) => {
  res.send('Hello User!')
})

app.listen(port, () => {
  console.log(`Server running on port ${port}`)
})
```

.env

PORT=4000

.gitignore

node_modules

.env

PART 3 — Run Locally

```
node index.js  
# OR  
npx nodemon index.js
```

Open browser: <http://localhost:4000>

PART 4 — Initialize Git

```
git init
```

Configure Git (once only)

```
git config --global user.name "Hasnain"  
git config --global user.email "hasnainvadaliwala@gmail.com"
```

PART 5 — Track Files & Commit

```
git status
git add .
git commit -m "Initial backend setup"
```

PART 6 — Create GitHub Repo

On GitHub: • New Repository • Name: back_ptc_1 • Choose Public/Private • Do NOT add README

PART 7 — Connect Local to GitHub

```
git remote add origin https://github.com/hasnainvadaliwala/back_ptc_1.git
git remote -v
git push -u origin main
```

PART 8 — Everyday Git Workflow

```
git status
git add .
git commit -m "updated routes"
git push
```

PART 9 — Deploy on Render

Render steps: 1. New Web Service 2. Connect GitHub 3. Select repo 4. Build: npm install 5. Start: node index.js

Add Environment Variable in Render dashboard:

PORT=10000

Why process.env.PORT || 3000 ?

It means: use platform-provided PORT if available, otherwise fall back to 3000 locally.

Overall Flow

Create project → Push to GitHub → Render pulls repo → installs dependencies → assigns PORT → Node listens → App goes live.